In mass spectrometry, a sample's molecules are ionized in a vacuum and then exposed to a uniform electric field created by a high voltage plate in the acceleration chamber. The electric field accelerates the ions until they arrive at the next section of the device, designated as the separation chamber. In this section, the drifting ions are sorted by their mass-to-charge ratio (m/q).

The separation chamber in a time-of-flight mass spectrometer (TOF-MS) is linear and has no electric or magnetic fields. The ions travel at a constant velocity through the chamber until they reach the detector. The time it takes for an ion to reach the detector depends on its m/q ratio.

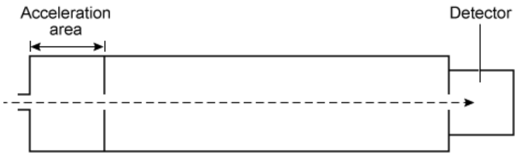


Figure 1 Time-of-flight mass spectrometer

A magnetic sector mass spectrometer (MS-MS) has a curved separation chamber in which a magnetic field is applied. The magnetic field exerts a centripetal force on drifting ions, bending their trajectories into curved paths. The radius of the curvature depends on the ion's m/q.

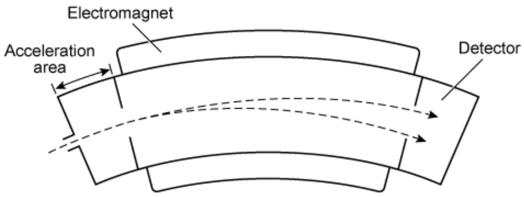


Figure 2 Magnetic sector mass spectrometer

The centripetal force F acting on a particle can be determined from its mass m, its velocity v, and the radius r of its curved path:

$$F = \frac{mv^2}{r}$$

Equation 1

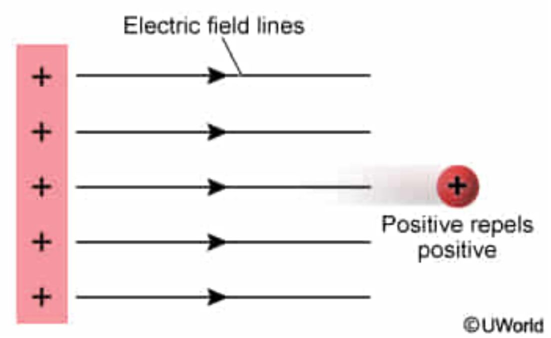
Which of the following best describes the high voltage plate if it accelerates positive ions away from it?

- A. It is positively charged, and its electric field lines point away from it. [72%]
- B. It is positively charged, and its electric field lines point toward it. [11%]
- C. It is negatively charged, and its electric field lines point away from it. [7%]
- D. It is negatively charged, and its electric field lines point toward it. [7%]

Correct

Correct
73% answered correctly

Explanation:



For the positive ion to be accelerated away from the plate, the charge on the plate needs to repel the ion. Charges with the **same sign repel** each other, and charges with opposite signs attract. Therefore, the accelerating plate needs to be **positively charged** to repel and accelerate positive ions away from it.

Electric field lines are used to denote the direction that a positive charge would be accelerated in an electric field. Because positive charges repel positive charges, electric field lines point outward from positive charges and point toward negative charges **(Choice B)**. Because the accelerating plate is positively charged, its electric field lines nearby must be **pointing away** from it, as shown in the figure.

(Choices C and D) If the plate were negatively charged, it would attract positive ions and prevent them from moving past the acceleration chamber. In addition, electric field lines would point toward the negatively charged plate.

Educational objective:
Electric charges with the same sign repel each other, and charges with opposite signs attract. Electric field lines point outward from positive charges and toward negative charges. Therefore, a positively charged plate repels positively charged ions, and the electric field lines point away from a positively charged plate.

Time spent: 0 sec	**QID**:400699
Subject: Physics	**System:** Electrostatics and Circuits

In mass spectrometry, a sample's molecules are ionized in a vacuum and then exposed to a uniform electric field created by a high voltage plate in the acceleration chamber. The electric field accelerates the ions until they arrive at the next section of the device, designated as the separation chamber. In this section, the drifting ions are sorted by their mass-to-charge ratio (*m*/*q*).

The separation chamber in a time-of-flight mass spectrometer (TOF-MS) is linear and has no electric or magnetic fields. The ions travel at a constant velocity through the chamber until they reach the detector. The time it takes for an ion to reach the detector depends on its *m*/*q* ratio.

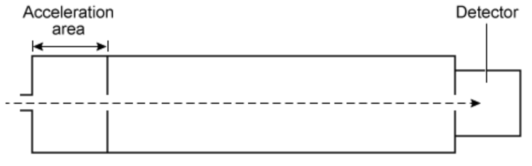


Figure 1 Time-of-flight mass spectrometer

A magnetic sector mass spectrometer (MS-MS) has a curved separation chamber in which a magnetic field is applied. The magnetic field exerts a centripetal force on drifting ions, bending their trajectories into curved paths. The radius of the curvature depends on the ion's *m*/*q*.

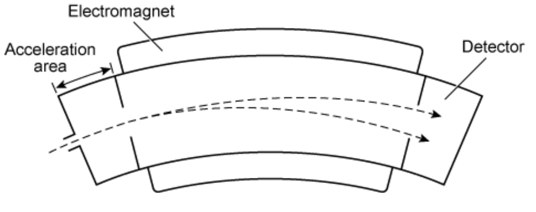


Figure 2 Magnetic sector mass spectrometer

The centripetal force *F* acting on a particle can be determined from its mass *m*, its velocity *v*, and the radius *r* of its curved path:

$$F = \frac{mv^2}{r}$$

Equation 1

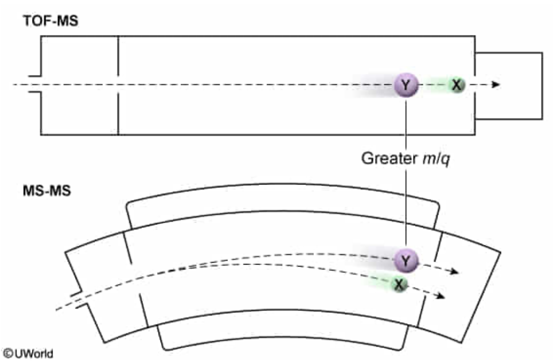
When ions **X** and **Y** are analyzed in a TOF-MS, ion **Y** took more time to reach the detector. Which ion's path will be observed having the smaller radius of curvature in MS-MS?

- A. Ion **X** [67%]
- B. Ion **Y** [26%]
- C. Both paths will have the same radius of curvature [3%]
- D. Neither path will curve in MS-MS [2%]

Correct

68% answered correctly

Explanation:



The TOF-MS is used to separate ions by the amount of time taken for the ions to travel a fixed distance. The ions are initially accelerated in a uniform electric field prior to entering the separation chamber. Ions of greater mass *m* and smaller charge *q* (ie, a higher *m*/*q* ratio) accelerate more slowly and take longer to reach the detector. Because ion **Y** is observed taking more time to reach the detector, **ion Y has a higher *m*/*q*ratio** and ion **X** has a lower *m*/*q* ratio.

In MS-MS, the force *F* exerted by the magnetic field on moving charges is directly proportional to their charge *q*. Because this force is centripetal $\left(F = \frac{mv^2}{r}\right)$, the ions are separated into paths of different curvature. Consequently:

$$\frac{mv^2}{r} \propto q$$

Rearranging this equation shows that the radius is proportional to the *m*/*q* ratio:

$$r \propto \frac{m}{q}$$

Ions with a higher *m*/*q* ratio are less affected by the magnetic field and will curve less within the magnetic field (ie, will have a larger radius of curvature). As a result, the ion with a lower *m*/*q* ratio, **ion X**, will be observed having a **smaller radius of curvature**.

(Choice B) Because ion **Y** has a higher *m*/*q* ratio, it is less affected by the magnetic field and its path has a larger radius of curvature.

(Choice C) Because ion **X** and ion **Y** took different amounts of time to reach the TOF-MS detector, they must have different *m*/*q* ratios. Therefore, their paths will not have the same radius of curvature because the radius also depends on the *m*/*q* ratio of the ion.

(Choice D) Because both ions have a nonzero charge and velocity when they exit the TOF-MS, they will both also follow a curved path in MS-MS.

Educational objective:
Ions in TOF-MS are separated by their travel time to the detector, which is directly proportional to their *m*/*q* ratio. Ions in MS-MS are separated by their radius of curvature, which is directly proportional to their *m*/*q* ratio. Therefore, an ion that takes less time to reach the detector in TOF-MS has a smaller *m*/*q* ratio and a smaller radius of curvature in MS-MS.

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Time spent:0 sec
Subject:
Physics

QID:400700
System:
Electrostatics and Circuits

In mass spectrometry, a sample's molecules are ionized in a vacuum and then exposed to a uniform electric field created by a high voltage plate in the acceleration chamber. The electric field accelerates the ions until they arrive at the next section of the device, designated as the separation chamber. In this section, the drifting ions are sorted by their mass-to-charge ratio (m/q).

The separation chamber in a time-of-flight mass spectrometer (TOF-MS) is linear and has no electric or magnetic fields. The ions travel at a constant velocity through the chamber until they reach the detector. The time it takes for an ion to reach the detector depends on its m/q ratio.

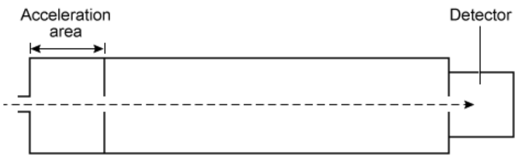


Figure 1 Time-of-flight mass spectrometer

A magnetic sector mass spectrometer (MS-MS) has a curved separation chamber in which a magnetic field is applied. The magnetic field exerts a centripetal force on drifting ions, bending their trajectories into curved paths. The radius of the curvature depends on the ion's m/q.

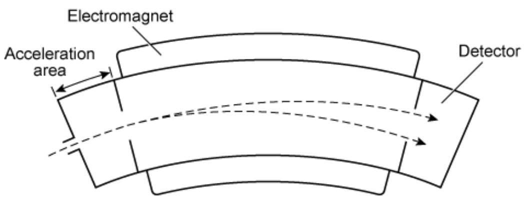


Figure 2 Magnetic sector mass spectrometer

The centripetal force F acting on a particle can be determined from its mass m, its velocity v, and the radius r of its curved path:

$$F = \frac{mv^2}{r}$$

Equation 1

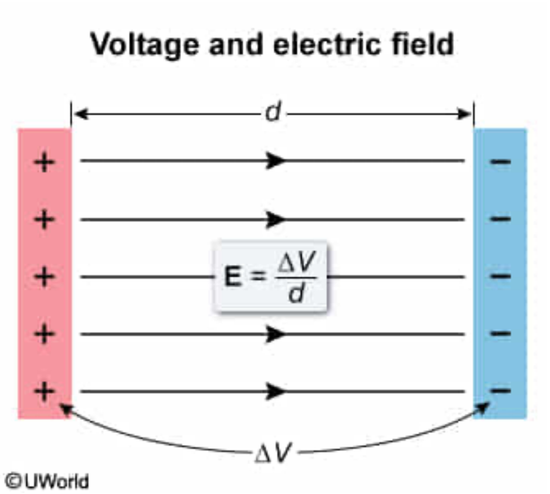
What is the magnitude of the electric field generated in the acceleration chamber if the potential difference is 3,000 V over a distance of 50 cm?

- A. 1.5 kN/C [19%]
- B. 3.0 kN/C [6%]
- C. 4.5 kN/C [7%]
- ✔ D. 6.0 kN/C [66%]

Correct
66% answered correctly

66% answered correctly

Explanation:



An **electric field** represents a space that exerts a force on charged particles. The magnitude of a uniform electric field *E* is the voltage difference *V* divided by the distance *d* of the field:

$$E = \frac{\Delta V}{d}$$

The magnitude of an electric field has SI units of **newtons per coulomb** (N/C), which is equivalent to **volts per meter** (V/m). Therefore, the given distance is converted from centimeters to meters (100 cm = 1 m):

$$d = 50\,\text{cm}\left(\frac{1\,\text{m}}{100\,\text{cm}}\right) = 0.5\,\text{m}$$

Using the given voltage (3,000 V) and the above distance, the magnitude of the electric field is:

$$E = \frac{3{,}000\,\text{V}}{0.5\,\text{m}} = 6{,}000\,\text{V/m}$$

This value is equal to 6,000 N/C, or 6.0 kN/C.

Educational objective:
An electric field accelerates charged particles, and its magnitude has SI units of newtons per coulomb (N/C), which are equivalent to volts per meter (V/m). The magnitude of a uniform electric field is calculated as its voltage divided by its distance: $E = \frac{\Delta V}{d}$.

Time spent:0 sec
Subject:
Physics

QID:400701
System:
Electrostatics and Circuits

In mass spectrometry, a sample's molecules are ionized in a vacuum and then exposed to a uniform electric field created by a high voltage plate in the acceleration chamber. The electric field accelerates the ions until they arrive at the next section of the device, designated as the separation chamber. In this section, the drifting ions are sorted by their mass-to-charge ratio (m/q).

The separation chamber in a time-of-flight mass spectrometer (TOF-MS) is linear and has no electric or magnetic fields. The ions travel at a constant velocity through the chamber until they reach the detector. The time it takes for an ion to reach the detector depends on its m/q ratio.

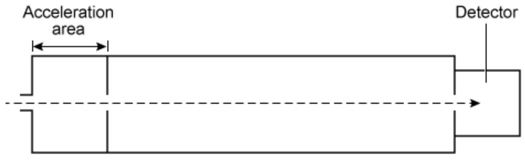


Figure 1 Time-of-flight mass spectrometer

A magnetic sector mass spectrometer (MS-MS) has a curved separation chamber in which a magnetic field is applied. The magnetic field exerts a centripetal force on drifting ions, bending their trajectories into curved paths. The radius of the curvature depends on the ion's m/q.

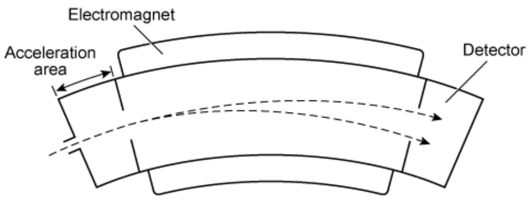


Figure 2 Magnetic sector mass spectrometer

The centripetal force F acting on a particle can be determined from its mass m, its velocity v, and the radius r of its curved path:

$$F = \frac{mv^2}{r}$$

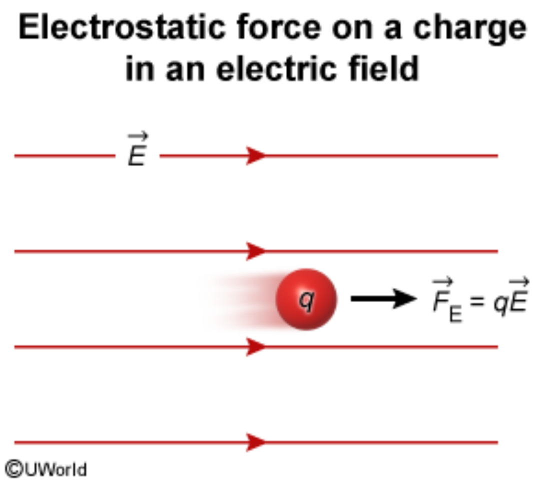
Equation 1

What is the force felt by a doubly ionized particle in a 2,000 N/C electric field? (Note: The charge of an electron is $e = 1.6 \times 10^{-19}$ C.)

- A. 1.6×10^{-22} N [2%]
- B. 3.2×10^{-22} N [7%]
- C. 3.2×10^{-16} N [41%]
- ✔ D. 6.4×10^{-16} N [48%]

Correct
49% answered correctly

Explanation:



Charged particles are accelerated in electric fields by a force known as the electric Lorentz force. The **force exerted on a particle in a uniform electric field** is equal to the product of the particle's charge *q* and the electric field strength *E*:

$$F = \text{qE}$$

The charge of a **doubly ionized particle** is twice the charge of an electron (1.6×10^{-19} C):

$$q = 2\left(1.6 \times 10^{-19}\ \text{C}\right) = 3.2 \times 10^{-19}\ \text{C}$$

Using the above charge as *q* and the given electric field strength of 2,000 N/C as *E*, the force exerted on the particle is:

$$F = \left(3.2 \times 10^{-19}\ \text{C}\right)\left(2{,}000\ \frac{\text{N}}{\text{C}}\right) = 6.4 \times 10^{-16}\ \text{N}$$

(Choice C) 3.2×10^{-16} N is the force exerted on a singly ionized molecule. However, the question asks for the force exerted on a doubly ionized molecule, which has twice the charge and therefore experiences twice the force.

Educational objective:
The force exerted on a particle in a uniform electric field is the product of its charge and the magnitude of the electric field: *F* = *qE*.

Time spent:0 sec
Subject:
Physics

QID:400702
System:
Electrostatics and Circuits

In mass spectrometry, a sample's molecules are ionized in a vacuum and then exposed to a uniform electric field created by a high voltage plate in the acceleration chamber. The electric field accelerates the ions until they arrive at the next section of the device, designated as the separation chamber. In this section, the drifting ions are sorted by their mass-to-charge ratio (m/q).

The separation chamber in a time-of-flight mass spectrometer (TOF-MS) is linear and has no electric or magnetic fields. The ions travel at a constant velocity through the chamber until they reach the detector. The time it takes for an ion to reach the detector depends on its m/q ratio.

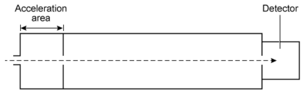


Figure 1 Time-of-flight mass spectrometer

A magnetic sector mass spectrometer (MS-MS) has a curved separation chamber in which a magnetic field is applied. The magnetic field exerts a centripetal force on drifting ions, bending their trajectories into curved paths. The radius of the curvature depends on the ion's m/q.

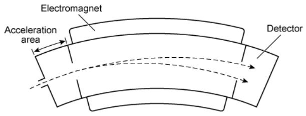


Figure 2 Magnetic sector mass spectrometer

The centripetal force F acting on a particle can be determined from its mass m, its velocity v, and the radius r of its curved path:

$$F = \frac{mv^2}{r}$$

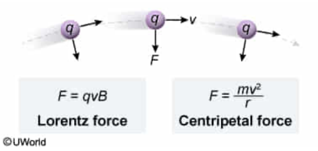
Equation 1

Which expression gives the radius of curvature in a magnetic field B for the path of an ion with a mass of m, charge of q, and velocity of v?

- A. $r = \left(\frac{m}{q}\right)\left(\frac{v^2}{B}\right)$ [34%]
- ✔ B. $r = \left(\frac{m}{q}\right)\left(\frac{v}{B}\right)$ [54%]
- C. $r = \left(\frac{q}{m}\right)\left(\frac{v^2}{B}\right)$ [7%]
- D. $r = \left(\frac{q}{m}\right)\left(\frac{v}{B}\right)$ [2%]

Correct
54% answered correctly

Explanation:



According to the passage, the **centripetal force** F exerted on an ion in MS-MS is related to the ion's mass m, its velocity v, and the radius r of its curved path (Equation 1):

$$F = \frac{mv^2}{r}$$

This force is a result of the **force exerted on an ion in a magnetic field**, known as the magnetic **Lorentz force**. It is calculated as the product of the ion's charge q, its velocity v, and the magnitude of the magnetic field B:

$$F = \text{qvB}$$

Because the force due to the magnetic field is the only force exerted on the ion, the centripetal force is equal to the Lorentz force:

$$\frac{mv^2}{r} = \text{qvB}$$

This equation can be rearranged to solve for the radius of curvature of the ion's path:

$$r = \frac{mv^2}{\text{qvB}} = \frac{\text{mv}}{\text{qB}}$$

This equation shows that the ion's path has a radius of curvature directly proportional to the ion's m/q ratio. In addition, the radius increases with the ion's velocity and decreases with the strength of the magnetic field.

(Choices C and D) These equations incorrectly state that r is proportional to q/m instead of the m/q ratio.

Educational objective:
The force exerted on a moving particle in a magnetic field is known as the magnetic Lorentz force, and it is equal to the product of the particle's charge, velocity, and strength of the magnetic field: $F = qvB$. This force is the centripetal force that curves the path of ions in MS-MS. Therefore, an expression for the radius of curvature can be determined by equating the Lorentz force and the centripetal force.

Time spent: 0 sec
Subject: Physics
QID: 400703
System: Electrostatics and Circuits

In mass spectrometry, a sample's molecules are ionized in a vacuum and then exposed to a uniform electric field created by a high voltage plate in the acceleration chamber. The electric field accelerates the ions until they arrive at the next section of the device, designated as the separation chamber. In this section, the drifting ions are sorted by their mass-to-charge ratio (*m*/*q*).

The separation chamber in a time-of-flight mass spectrometer (TOF-MS) is linear and has no electric or magnetic fields. The ions travel at a constant velocity through the chamber until they reach the detector. The time it takes for an ion to reach the detector depends on its *m*/*q* ratio.

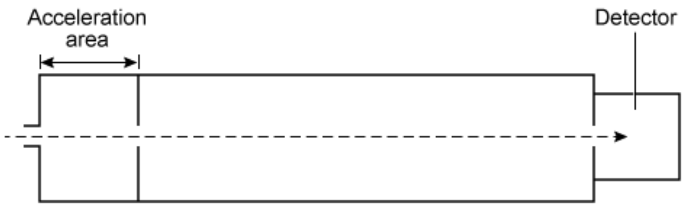


Figure 1 Time-of-flight mass spectrometer

A magnetic sector mass spectrometer (MS-MS) has a curved separation chamber in which a magnetic field is applied. The magnetic field exerts a centripetal force on drifting ions, bending their trajectories into curved paths. The radius of the curvature depends on the ion's *m*/*q*.

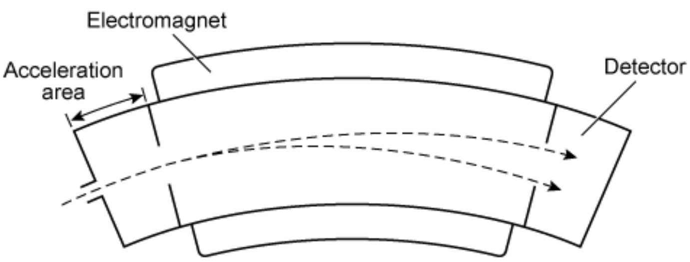


Figure 2 Magnetic sector mass spectrometer

The centripetal force *F* acting on a particle can be determined from its mass *m*, its velocity *v*, and the radius *r* of its curved path:

$$F = \frac{mv^2}{r}$$

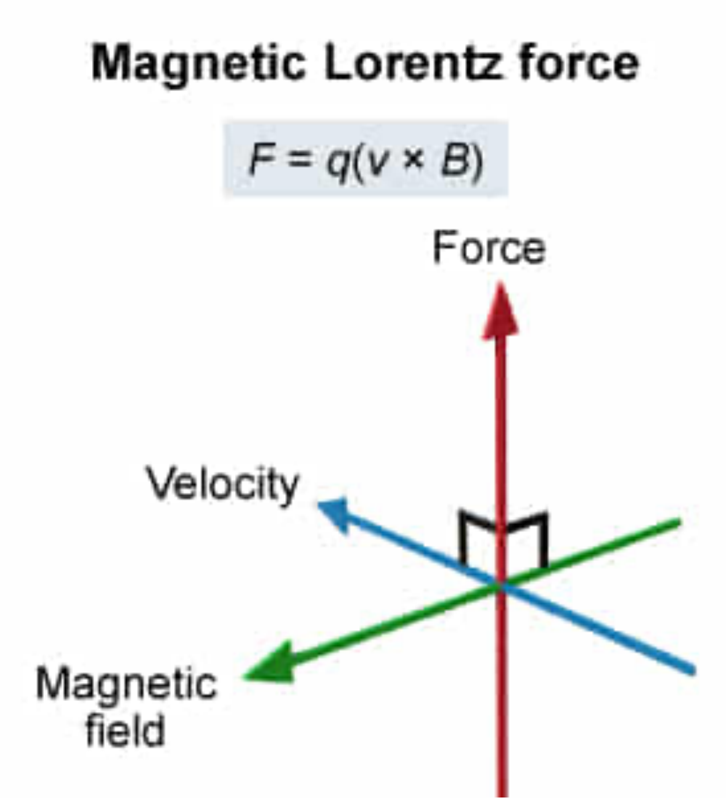
Equation 1

In the MS-MS separation chamber, the direction of the magnetic force on a moving ion is:

- ✔ ◉ A. perpendicular to both the ion's velocity and the direction of the magnetic field. [58%]
- ○ B. perpendicular to the ion's velocity and parallel to the direction of the magnetic field. [25%]
- ○ C. parallel to both the ion's velocity and the direction of the magnetic field. [4%]
- ○ D. parallel to the ion's velocity and perpendicular to the direction of the magnetic field. [12%]

Correct
58% answered correctly

Explanation:



The **Lorentz force** is the force exerted on a moving charge in the presence of an electric field *E* and a magnetic field *B*. Electric and magnetic fields are known as **vector fields**; both quantities include magnitude and direction. The Lorentz force for a particle with a charge *q* and velocity *v* in an electric field *E* and magnetic field *B* is:

$$\vec{F} = q\left(\vec{E} + \vec{v} \times \vec{B}\right)$$

The quantities *F*, *E*, *v*, and *B* are denoted as vector quantities. Because there is no electric field (*E* = 0) in the MS-MS separation chamber, the Lorentz force is simplified for the presence of a magnetic field only:

$$\vec{F} = q\left(\vec{v} \times \vec{B}\right)$$

The quantity $\vec{v} \times \vec{B}$ is the **cross product** of the vectors $\vec{v}$ and $\vec{B}$. The cross product of two vectors is a vector perpendicular to both vectors. Therefore, the direction of the **magnetic force is perpendicular to both** the ion's velocity *v* and the direction of the magnetic field *B*. The specific direction of a cross-product vector can be determined through the right-hand rule.

(Choice B) The force due to an *electric* field is parallel to the direction to the electric field, but the force due to a magnetic field is perpendicular to its field.

(Choices C and D) If the magnetic force were parallel to the ion's velocity, the ion would accelerate linearly and not be forced into a curved trajectory.

Educational objective:
The Lorentz force equation is used to determine the force exerted on a charge in the presence of an electric field and a magnetic field. The force exerted on a moving charge due to a magnetic field is perpendicular to both the ion's velocity and the direction of the magnetic field.

Time spent: 0 sec	**QID**:400704
Subject:	**System:**
Physics	Electrostatics and Circuits